

STOR 155.010 Lecture 09

Basic Probability

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Fall 2026
UNC Chapel Hill

15 September 2026

From Data to Models: Why Probability?

- So far this semester, we've focused on **statistics**: given data, how do we summarize it, visualize it, and draw conclusions from it?
- But where does data actually *come from*? In many settings, we imagine data as being generated by some underlying random process, or **model**.
- **Probability** is the study of these underlying models; it tells us what outcomes to *expect*, before we ever collect data.
- Two sides of the same coin:
 - **Probability**: Model $\longrightarrow$ Data. If we know the rules of the game, what data should we expect to see?
 - **Statistics**: Data $\longrightarrow$ Model. Given data we've observed, what can we infer about the underlying rules?

Why This Matters for Statistics

- Every statistical method we have used so far (mean, standard deviation, correlation, sampling, etc.) implicitly assumes some random process generated the data.
- To reason rigorously about *how confident* we should be in a conclusion from data (e.g. “is this difference real, or just chance?”), we need probability.
- Over the next few lectures, we’ll build the language of probability from the ground up: random experiments, outcomes, events, and rules for combining them.
- This is the foundation for everything that follows: random variables, distributions, and eventually statistical inference (confidence intervals, hypothesis tests).

Random Experiments

- **Random experiment:** A repeatable procedure/activity where all possible outcomes are known beforehand, but the exact outcome cannot be predicted with certainty.
 - *Example.* Rolling a die, tossing 2 coins, drawing 4 cards, etc.
- **Sample Space (Ω):** Set of all possible outcomes of an experiment.
- **Elementary outcome ($\omega \in \Omega$):** Result of a random experiment.
- **Event ($A \subseteq \Omega$):** A subset of the sample space.
 - The result of rolling a die being an even number.
 - Getting atleast one head in two tosses of a coin.

Random Experiments: Examples

- **Tossing two coins:**

- $\Omega = \{HH, HT, TH, TT\}$
- Event $A = \text{"at least one Head"} = \{HH, HT, TH\}$

- **Rolling a die:**

- $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Event $B = \text{"an even number"} = \{2, 4, 6\}$

- **Picking 2 people from a group of 5 (named 1, 2, 3, 4, 5):**

- $\Omega = \{12, 13, 14, 15, 23, 24, 25, 34, 35, 45\}$
- Event $C = \text{"person 1 is picked"} = \{12, 13, 14, 15\}$

Event Operations

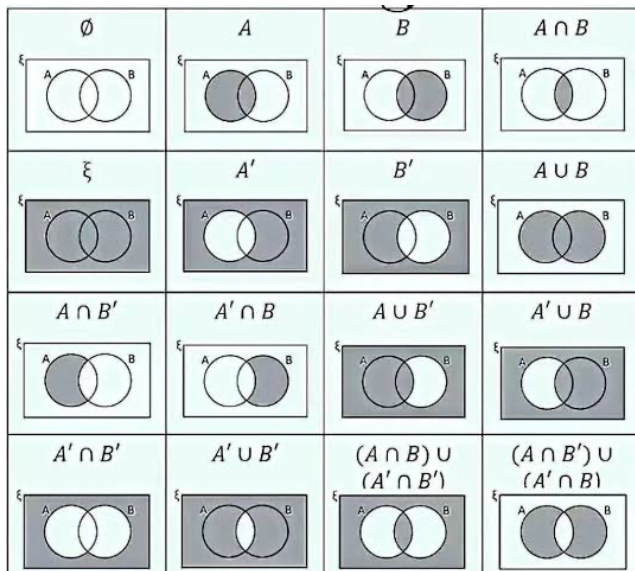
Let $U, V \subseteq \Omega$ be two events.

- **Union:** $U \cup V$ is the event either U or V (or both) occur.
- **Intersection:** $U \cap V$ is the event both U and V occur.
- **Complement:** U^c is the event U does not occur.
- **Set difference:** $U \setminus V$ is the event U occurs but not V .
- **Disjoint:** Say U and V are disjoint if $U \cap V = \emptyset$, i.e. they cannot both occur at the same time.
 - If U occurs, then V cannot and vice versa.

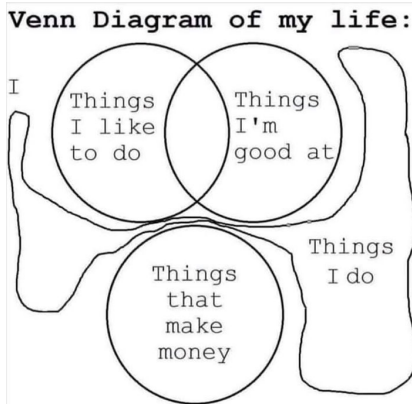
Event Operations: Examples

- **Tossing two coins:** $U = \text{"at least one Head"} = \{HH, HT, TH\}$,
 $V = \text{"at least one Tail"} = \{HT, TH, TT\}$.
 - $U \cup V = \{HH, HT, TH, TT\} = \Omega$
 - $U \cap V = \{HT, TH\}$
 - $U^c = \{TT\}$
- **Rolling a die:** $U = \text{"even"} = \{2, 4, 6\}$, $V = \text{"greater than 4"} = \{5, 6\}$.
 - $U \cap V = \{6\}$
 - $U \setminus V = \{2, 4\}$ and $V \setminus U = \{5\}$
- **Picking 2 from 1, 2, 3, 4, 5:** $U = \text{"1 is picked"}$, $V = \text{"2 is picked"}$.
 - $U \cap V = \{12\}$ (both 1 and 2 picked)
 - Are U and V disjoint? **No**, both can happen at once (pair 12).

Venn Diagram



Venn Diagram Examples



Practice Example

Problem: A class of 30 students is composed of 18 males and 12 females. Out of these, 10 males and 7 females have brown eyes.

- Event G : A randomly selected student is male.
- Event H : A randomly selected student has brown eyes.
- Event I : A randomly selected student is female.

Questions:

- 1 What does the event $G \cap H$ represent? How many elements does it have?
- 2 What does the event G^c ? How many elements does it have?
- 3 What does the event $H \setminus I$ represent, and how many elements does it have?

- Informally, the probability of an event is the chance of occurrence of the event.
- For an event A , denote the probability of A by $\mathbb{P}(A)$.
 - $0 \leq \mathbb{P}(A) \leq 1$ for all $A \subseteq \Omega$.
 - $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$.
 - If A and B are disjoint events, then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.
- Two important properties:
 - $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
 - $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
- Events A and B are **mutually exclusive** if $\mathbb{P}(A \cap B) = 0$.

Probability: Examples

- **Tossing two coins** (all 4 outcomes equally likely, $\mathbb{P}(\omega) = 1/4$ each):
 - $\mathbb{P}(\text{at least one Head}) = \mathbb{P}(\{HH, HT, TH\}) = 3/4$
 - Using the complement: $\mathbb{P}(\text{no Heads}) = \mathbb{P}(\{TT\}) = 1/4$, so $\mathbb{P}(\text{at least one Head}) = 1 - 1/4 = 3/4$.
- **Rolling a die** (each face equally likely, $\mathbb{P}(\omega) = 1/6$):
 - $\mathbb{P}(\text{even}) = \mathbb{P}(\{2, 4, 6\}) = 3/6 = 1/2$
 - $\mathbb{P}(\text{even} \cup \text{greater than 4}) = \mathbb{P}(\{2, 4, 6\}) + \mathbb{P}(\{5, 6\}) - \mathbb{P}(\{6\}) = 3/6 + 2/6 - 1/6 = 4/6 = 2/3$
- **Picking 2 from 1–5** (all 10 pairs equally likely, $\mathbb{P}(\omega) = 1/10$):
 - $\mathbb{P}(1 \text{ is picked}) = \mathbb{P}(\{12, 13, 14, 15\}) = 4/10 = 2/5$
 - $\mathbb{P}(1 \text{ is picked} \cap 2 \text{ is picked}) = \mathbb{P}(\{12\}) = 1/10$